# Infosys System Engineer: Mock Test — Paper 5

**Total questions:** 60 | **Duration:** 120 minutes | **Sectional cut-offs apply**

| Section | Questions | Time |
|---|---|---|
| 1. Reasoning Ability | Q1–15 | 25 min |
| 2. Technical Ability (Quant) | Q16–25 | 35 min |
| 3. Verbal Ability | Q26–45 | 20 min |
| 4. Pseudo Code | Q46–50 | 10 min |
| 5. Numeric Puzzle Solving | Q51–54 | 10 min |
| 6. English Grammar | Q55–59 | 10 min |
| 7. English Writing | Q60 | 10 min |

## SECTION 1: Reasoning Ability (15 Q, 25 min)

**Q1.** Statements: All toys are games. All games are activities. Some crafts are not activities. No craft is a hobby.

Conclusions: I. All toys are activities. II. Some crafts are not hobbies. III. No toy is a craft.

(a) Only I follows (b) I and II follow (c) I, II and III follow (d) None follows

**Q2.** Statements: All researchers are scholars. No scholar is lazy. Some lazy people are students. All students are young.

Conclusions: I. No researcher is lazy. II. Some students are not scholars. III. Some young people are lazy.

(a) Only I and II follow (b) Only II and III follow (c) All follow (d) None follows

**Q3.** *Data Sufficiency.* If x and y are positive integers, what is the remainder when x is divided by y?

I. $x = 5y + 3$

II. $y = 7$

(a) Statement I alone is sufficient, but II alone is not

(b) Statement II alone is sufficient, but I alone is not

(c) Both statements together are sufficient, but neither alone is

(d) Even both statements together are not sufficient

**Q4.** *Data Sufficiency.* In how many days can A and B complete a job together?

I. A is twice as efficient as B.

II. A alone can complete the job in 12 days.

(a) Statement I alone is sufficient, but II alone is not

(b) Statement II alone is sufficient, but I alone is not

(c) Both statements together are sufficient, but neither alone is

(d) Even both statements together are not sufficient

**Q5.** In a code, letters in odd positions of a word are shifted 2 places forward in the alphabet, and letters in even positions are shifted 1 place forward. Using this rule, SYSTEM is coded as UZUUGN. What is the code for SERVER?

(a) UFTWGS (b) UFTWFS (c) UGTWGS (d) UFUWGS

**Q6.** In a code: "data packet loss" = "zt rp qm", "loss causes retransmission" = "qm xy lp", "retransmission increases latency" = "lp wv ou", "latency affects data" = "ou nk zt", "packet causes latency" = "rp xy ou". What is the code for "increases affects packet"?

(a) wv nk rp (b) wv nk qm (c) ou nk rp (d) wv xy rp

**Q7.** A rectangular cuboid is cut by 2 parallel cuts along its length, 1 cut along its breadth, and 3 cuts along its height (each cut perpendicular to that dimension), producing smaller cuboids from an originally fully-painted surface. How many of the resulting small cuboids have exactly 1 painted face?

(a) 4 (b) 6 (c) 8 (d) 2

**Q8.** How many rectangles (of all sizes, including squares) are there in a 4×4 grid of unit squares?

(a) 100 (b) 90 (c) 81 (d) 110

**Q9.** P and Q are a married couple. R is the son of P and Q. S is the daughter of P and Q. R is married to T. U is the son of R and T. V is the daughter of S. How is U related to V?

(a) Cousins (b) Siblings (c) Nephew and Niece (d) Uncle and Niece

**Q10.** If A + B means A is the father of B, A − B means A is the mother of B, A × B means A is the son of B, and A ÷ B means A is the wife of B, which expression shows that M is the father-in-law of N (N being male)?

(a) M + Q ÷ N (b) M ÷ Q + N (c) Q + M ÷ N (d) M × Q ÷ N

**Q11.** A man starts from point O and walks 10 m North-East, then 10 m North-West, then 10 m South-West. How far is he from O, and in which direction?

(a) 10 m, North-West (b) 10 m, South-West (c) 14.14 m, North-West (d) 5 m, North-West

**Q12.** A man is initially facing a bearing of 040°. After making exactly 2 equal clockwise turns, he ends up facing a bearing of 310°. What was the magnitude of each turn?

(a) 135° (b) 142.5° (c) 120° (d) 150°

**Q13.** Find the next term: 5, 14, 41, 122, 365, ?

(a) 1094 (b) 1095 (c) 1093 (d) 1096

**Q14.** Find the next term in this 7-term series: 4, 9, 7, 16, 10, 25, 13, ?

(a) 36 (b) 16 (c) 49 (d) 30

**Q15.** Find the next term: D, H, M, S, Z, ?

(a) H (b) G (c) I (d) F

## SECTION 2: Technical Ability, Quantitative (10 Q, 35 min)

**Q16.** A committee of 5 is to be formed from 6 men and 5 women. A specific man X must be included, a specific woman Y must be excluded, and the committee must include at least 1 woman. In how many ways can this be done?

(a) 121 (b) 126 (c) 120 (d) 115

**Q17.** In how many ways can the 8 distinct letters of COMPUTER be arranged such that C and T have exactly 2 letters between them (in either order)?

(a) 7200 (b) 6000 (c) 8400 (d) 5040

**Q18.** Two tickets are drawn one after another, without replacement, from a box containing tickets numbered 1 to 10. What is the probability that the number on the second ticket is greater than the number on the first?

(a) 1/2 (b) 9/20 (c) 11/20 (d) 2/5

**Q19.** A mini-deck has 16 cards: 4 suits with 4 cards each. 4 cards are drawn at random. What is the probability that at least 2 of them share the same suit?

(a) 391/455 (b) 64/455 (c) 327/455 (d) 201/455

**Q20.** A shopkeeper runs a "Buy 3, Get 1 Free" offer on an item marked at ₹200 (costing him ₹120 each). A customer buys 4 items under this offer, paying for 3. What is the shopkeeper's profit percentage, based on his cost for all 4 items given?

(a) 25% (b) 20% (c) 30% (d) 15%

**Q21.** A, B and C can individually complete a work in 15, 20 and 30 days. They work together, and the total payment of ₹9,000 is distributed in proportion to each person's work contribution. How much should C be paid?

(a) ₹2,000 (b) ₹1,800 (c) ₹2,200 (d) ₹2,500

**Q22.** Two runners start from the same point on a 600 m circular track at the same time, in opposite directions, at 8 m/s and 10 m/s. After how many seconds will they both simultaneously be back at the starting point for the first time?

(a) 300 s (b) 200 s (c) 150 s (d) 600 s

**Q23.** A container has 80 litres of pure milk. 20 litres of the mixture is removed and replaced with water; this process is repeated once more. How much pure milk remains?

(a) 45 litres (b) 50 litres (c) 40 litres (d) 48 litres

**Q24.** At a certain rate of annual compound interest, a sum doubles itself in 5 years. In how many years will it become 8 times itself, at the same rate?

(a) 15 years (b) 20 years (c) 10 years (d) 25 years

**Q25.** ₹20,000 is borrowed at 10% per annum compound interest (annual compounding). ₹11,000 is repaid at the end of year 1. What amount must be paid at the end of year 2 to clear the entire debt?

(a) ₹12,100 (b) ₹12,000 (c) ₹11,500 (d) ₹13,200

## SECTION 3: Verbal Ability (20 Q, 20 min)

**Read the passage and answer Q26–30.**

> *Rate limiting is often presented as a purely defensive mechanism, a way to shield backend services from abusive or accidental traffic spikes, but this framing understates its role as a tool for communicating capacity contracts between services. A token-bucket limiter that rejects requests once a quota is exhausted is, in effect, declaring an explicit promise about how much load a downstream service can absorb, and client-side code written against that promise can make informed decisions about retry timing, backoff, and degraded functionality rather than failing unpredictably. The trouble arises when limits are set reactively, in response to an incident, rather than derived from actual capacity testing; a*

> *limit chosen under pressure to stop an ongoing outage often reflects the conditions of that specific incident rather than the service's true sustainable throughput, and may be far too conservative once normal operation resumes, quietly capping legitimate growth long after the original emergency has been forgotten. Treating rate limits as living documentation of measured capacity, revisited whenever the underlying service's performance characteristics change, avoids both failure modes: the limiter protects the service without becoming an invisible ceiling on its own success.*

**Q26.** The passage's central argument is that rate limits should be understood primarily as:
(a) A purely defensive mechanism against malicious traffic
(b) An explicit, measured capacity contract that client code can plan around, not just a defensive wall
(c) A permanent configuration that should never be revisited
(d) A substitute for proper capacity testing

**Q27.** According to the passage, limits set reactively during an incident risk becoming:
(a) Too generous, allowing further outages
(b) An invisible ceiling that quietly caps legitimate growth after the emergency has passed
(c) Immediately replaced by automated systems
(d) Irrelevant to client-side retry logic

**Q28.** The phrase "living documentation of measured capacity" implies that rate limits should be:
(a) Set once and never changed
(b) Periodically revisited as the service's performance characteristics evolve
(c) Removed once a service proves reliable
(d) Determined solely by client demand

**Q29.** Which of the following would the author most likely agree with?
(a) A well-chosen rate limit communicates useful information to the systems that call it, beyond just blocking excess traffic
(b) Rate limits should always be set as conservatively as possible
(c) Capacity testing is unnecessary if a rate limiter is in place
(d) Client-side retry logic is unrelated to how a rate limit is set

**Q30.** Based on the passage, what is the main risk of NOT revisiting a reactively-set limit?
(a) The service becomes vulnerable to outages again
(b) The limit may stay more conservative than necessary and suppress legitimate growth
(c) Client code will stop retrying altogether
(d) The token-bucket algorithm will stop functioning

**Sentence Correction (Q31–35).**

**Q31.** If the migration **was** properly tested, the system **would not be** experiencing intermittent failures right now.
(a) No improvement (b) had been, would not be (c) were, will not be (d) has been, would not have been

**Q32.** The lead engineer said that the deployment **will** take place the following week, as the testing **has** not yet been completed at that time.
(a) No improvement (b) would, had (c) would, has (d) will, had

**Q33.** The team had to account **for the discrepancy, which they eventually traced to** a configuration error introduced **by** mistake during the last deployment.

(a) No improvement (b) of, to, in (c) with, into, by (d) for, into, with

**Q34.** The new policy requires all employees **to attending** mandatory security training within thirty days of joining, failing which their system access will be automatically suspended.

(a) No improvement (b) attending (c) to attend (d) attend

**Q35.** The outage **was caused due to** a misconfigured load balancer that had been deployed without peer review.

(a) No improvement (b) was caused by (c) was caused because of (d) caused due to

**Para-jumbles (Q36–38).**

**Q36.**

P: This made it difficult to reason about which version of a shared object any given thread would observe.

Q: Early concurrent programming models allowed any thread to read or write shared memory at any time.

R: Memory models that formally define visibility and ordering guarantees were developed to resolve this ambiguity.

S: Without explicit synchronisation, two threads could see updates to the same variable in different orders.

(a) QSPR (b) QPSR (c) SQPR (d) QSRP

**Q37.**

P: A cache invalidated too eagerly provides little performance benefit, defeating its own purpose.

Q: Cache invalidation is frequently cited as one of the two genuinely hard problems in computer science.

R: One invalidated too late serves stale data that may silently corrupt downstream computations.

S: The difficulty lies in striking a balance between staleness and overhead that rarely has a universally correct answer.

(a) QPRS (b) QRPS (c) QPSR (d) PQRS

**Q38.**

P: Engineers began instrumenting every service boundary to capture latency at each hop.

Q: A single slow request in a microservice architecture can be caused by any one of a dozen interacting services.

R: Distributed tracing tools emerged to stitch these individual measurements into one coherent timeline.

S: Without this visibility, a single slow endpoint could remain unexplained for weeks.

(a) QSPR (b) QPSR (c) SQPR (d) QSRP

**Vocabulary, Synonyms and Antonyms (Q39–45).**

**Q39.** The consultant's recommendations were **anodyne**, offering nothing that would upset any stakeholder. "Anodyne" means:

(a) controversial (b) inoffensive and bland (c) insightful (d) aggressive

**Q40.** The engineer's **terse** response left no room for follow-up questions. "Terse" means:

(a) elaborate (b) brief and to the point, almost curt (c) friendly (d) confusing

**Q41.** Fill in the blank: The postmortem was praised for being _____ rather than accusatory, focusing on systemic causes instead of individual blame.

(a) punitive (b) blameless (c) hostile (d) dismissive

**Q42.** Synonym of **INTRACTABLE**:

(a) manageable (b) difficult to control or solve (c) flexible (d) simple

**Q43.** Synonym of **PRECARIOUS**:

(a) stable (b) unstable or risky (c) secure (d) permanent

**Q44.** Antonym of **MERITORIOUS**:

(a) praiseworthy (b) commendable (c) blameworthy (d) deserving

**Q45.** Antonym of **VINDICATE**:

(a) justify (b) exonerate (c) incriminate (d) clear

## SECTION 4: Pseudo Code (5 Q, 10 min)

**Q46.** What is printed? (This computes the number of grid paths from one corner to another, moving only right or down.)

```
int f(int n, int m) {
    if (n == 0 || m == 0) return 1;
    return f(n-1, m) + f(n, m-1);
}
print(f(3, 2))
```

(a) 10 (b) 8 (c) 12 (d) 9

**Q47.** What is printed? (>>=right shift, <<=left shift, &=AND, ^=XOR)

```
x = 0;
for i = 0 to 7
    if (((45 >> i) & 1) == 1)
        x = x + (1 << (7 - i))
    end if
end for
y = x ^ 45
print(y)
```

(a) 153 (b) 180 (c) 135 (d) 171

**Q48.** This is a binary search for `target = 50` in a sorted array. How many comparisons are made before the loop ends (the target is not present)?

```
arr = [2, 5, 8, 12, 16, 23, 38, 45, 56, 72, 91]
low = 0; high = 10; target = 50; comparisons = 0
while (low <= high)
    mid = (low + high) / 2
    comparisons = comparisons + 1
    if (arr[mid] == target)
        break
    else if (arr[mid] < target)
        low = mid + 1
    else
        high = mid - 1
    end if
end while
print(comparisons)
```

(a) 4 (b) 3 (c) 5 (d) 6

**Q49.** This stack trace evaluates a postfix-style expression. What is printed?

```
push(6); push(2); push(3);
c = pop(); b = pop(); push(b + c);
d = pop(); e = pop(); push(e - d);
push(3); push(8); push(2);
f = pop(); g = pop(); push(g / f);
h = pop(); i = pop(); push(i + h);
j = pop(); k = pop(); push(k * j);
print(pop())
```

(a) 7 (b) 10 (c) 14 (d) 4

**Q50.** This is Kadane's algorithm for maximum subarray sum. What is printed?

```
arr = [-2, 3, -1, 5, -3, 4, -2, 6]
maxSoFar = arr[0]; maxEndingHere = arr[0]
for i = 1 to 7
    maxEndingHere = max(maxEndingHere + arr[i], arr[i])
    maxSoFar = max(maxSoFar, maxEndingHere)
end for
print(maxSoFar)
```

(a) 12 (b) 10 (c) 9 (d) 15

## SECTION 5: Numeric Puzzle Solving (4 Q, 10 min)

**Q51.** *Matchstick.* Five unit squares are arranged in a plus/cross pattern (one centre square with one square attached to each of its four sides), using 16 matchsticks in total. What is the minimum number of sticks that must be removed so that no complete square remains?

(a) 3 (b) 4 (c) 5 (d) 6

**Q52.** *Circular wheel.* Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.

Wheel 1: T = 8, R = 2, B = 6, L = 3, Centre = 41

Wheel 2: T = 10, R = 4, B = 5, L = 2, Centre = 65

Wheel 3: T = 12, R = 3, B = 9, L = 5, Centre = ?

(a) 85 (b) 81 (c) 90 (d) 78

**Q53.** *Matrix logic.* Find the missing number.

| | | |
|---|---|---|
| 3 | 7 | 13 |
| 7 | 12 | 19 |
| 13 | 19 | ? |

(a) 25 (b) 27 (c) 29 (d) 24

**Q54.** *Pattern.* Find the next term: 3, 9, 19, 33, 51, ?

(a) 73 (b) 69 (c) 77 (d) 71

## SECTION 6: English Grammar (5 Q, 10 min)

**Q55.** If the migration _____ properly tested, the system _____ experiencing intermittent failures right now.

(a) was, would not be (b) had been, would not be (c) were, will not be (d) has been, would not have been

**Q56.** The lead engineer said that the deployment _____ take place the following week, as the testing _____ not yet been completed at that time.

(a) would, had (b) will, has (c) would, has (d) will, had

**Q57.** The team had to account _____ the discrepancy, which they eventually traced _____ a configuration error introduced _____ mistake during the last deployment.

(a) for, to, by (b) of, to, in (c) with, into, by (d) for, into, with

**Q58.** *Error spotting.* "The new policy requires all employees to attending mandatory security training (A) / within thirty days of joining, (B) / failing which their system access (C) / will be automatically suspended. (D)"

(a) A (b) B (c) C (d) D

**Q59.** _____ United Nations was founded after _____ World War II, and _____ UN Security Council now includes _____ five permanent members.

(a) The, the, the, the (b) -, the, the, - (c) The, -, the, the (d) The, the, -, the

## SECTION 7: English Writing (1 Q, 10 min)

**Q60.** *Write a short technical essay/paragraph of 150–180 words.*

Your team is choosing between two caching strategies for a high-traffic API: a write-through cache (data written to cache and database simultaneously) versus a write-back cache (data written to cache first, then asynchronously to the database). Explain the key trade-offs between these approaches in terms of data consistency and performance, and recommend which strategy best suits (a) a financial transactions API, and (b) a social media "like" counter — justifying each recommendation briefly.

**Evaluation rubric (10 marks):** Technical accuracy of the trade-off explanation (3) | Soundness of both context-specific recommendations (3) | Structure and coherence (2) | Grammar and vocabulary (1) | Word limit adherence (1)

# ANSWER KEY WITH EXPLANATIONS

## Section 1: Reasoning

1. **(b)** Toys⊆games⊆activities, so all toys are activities (I follows). "Some crafts are not activities" confirms crafts exist, and since no craft is a hobby, those crafts are also not hobbies — "some crafts are not hobbies" follows (II). III is not guaranteed, since nothing links toys specifically to crafts.

2. **(c)** Researchers⊆scholars, and scholars∩lazy=∅, so researchers∩lazy=∅ (I). Some lazy are students, and since scholars∩lazy=∅, those lazy-students are not scholars — "some students are not scholars" (II). Those same lazy-students are also students, and all students are young, so those lazy-students are young — "some young people are lazy" (III). All three follow.

3. **(c)** I alone: $x=5y+3$ only gives remainder 3 if $y>3$; for small y (e.g., y=2, x=13, 13÷2 leaves remainder 1, not 3) it fails — insufficient alone. II alone: y=7 with x unknown is insufficient. Together: $x=5(7)+3=38$, and 38÷7 leaves remainder 3 (valid since $3<7$) — sufficient together.

4. **(c)** II alone gives only A's rate (1/12), with B's rate unknown. I alone gives only a ratio (2:1), with no absolute rate. Together: A's rate=1/12, B's rate=1/24 (half of A's), combined rate=1/12+1/24=3/24=1/8, so together they take 8 days — sufficient together.

5. **(a)** S(odd,+2)=U. E(even,+1)=F. R(odd,+2)=T. V(even,+1)=W. E(odd,+2)=G. R(even,+1)=S. Code: **UFTWGS**.

6. **(a)** Pairwise comparison of the 5 statements resolves every word: data=zt, packet=rp, loss=qm, causes=xy, retransmission=lp, increases=wv, latency=ou, affects=nk. So "increases affects packet" = **wv nk rp**.

7. **(a)** With 2 cuts in length (3 pieces), 1 cut in breadth (2 pieces), 3 cuts in height (4 pieces), every piece touches an outer breadth-face (since there's no "middle" breadth layer). A piece has exactly 1 painted face only if it is interior in BOTH length and height (contributing 0 from each) while contributing exactly 1 from breadth (which is unavoidable). Interior length positions = 1 (out of 3), interior height positions = 2 (out of 4), and both breadth pieces qualify: 1×2×2 = **4**.

8. **(a)** Choosing 2 of the 5 horizontal lines and 2 of the 5 vertical lines of a 4×4 grid gives a rectangle: $C(5,2)\times C(5,2) = 10\times10 =$ **100**.

9. **(a)** R and S are siblings (children of P and Q). U is R's son; V is S's daughter. Since their parents (R and S) are siblings, U and V are **cousins**.

10. **(a)** M+Q: M is father of Q. Q÷N: Q is wife of N. So M is father of N's wife — M is N's **father-in-law**.

11. **(a)** Decomposing each 10 m leg into East/North components (each at 45°, so ±5√2 in each axis): totals work out to East=−5√2, North=+5√2, giving a resultant of $\sqrt{[(5\sqrt2)^2+(5\sqrt2)^2]}=\sqrt{100}=$**10 m**, in the **North-West** direction.

12. **(a)** Total clockwise rotation needed = 310°−40° = 270°. Split equally over 2 turns: 270/2 = **135°** each (verified: 40°+135°=175°, 175°+135°=310° ✓).

13. **(a)** Each term follows $a(n) = 3\times a(n-1) - 1$: $122\times3-1=365$ ✓; next = $365\times3-1 =$ **1094**.

14. **(a)** This series interleaves two sequences: odd positions (4,7,10,13,...) increase by 3; even positions (9,16,25,...) are $3^2,4^2,5^2$. The 8th term (even position) continues the squares: $6^2 =$ **36**.

15. **(a)** Gaps increase by 1 each time: +4,+5,+6,+7; next gap +8. Z(26)+8=34, which wraps around (34−26=8) to **H**.

## Section 2: Quant

16. **(a)** With X fixed in and Y fixed out, the remaining pool is 5 men + 4 women = 9 people, from which 4 more members are chosen: C(9,4)=126. Subtract the invalid case where all 4 additional members are men (no woman at all): C(5,4)=5. Valid = 126−5 = **121**.

17. **(a)** Valid position-pairs with C and T exactly 3 apart: 5 pairs with C before T, plus 5 with T before C = 10 total. The remaining 6 distinct letters fill the remaining 6 positions in 6!=720 ways. Total = 10×720 = **7,200**.

18. **(a)** By symmetry, for any two distinct numbers drawn in some order, exactly half the time the second exceeds the first. Probability = **1/2** (confirmed via direct count: 45 favourable ordered outcomes out of 90).

19. **(a)** P(all 4 different suits) = (4×4×4×4)/C(16,4) = 256/1820 = 64/455. P(at least 2 same suit) = 1−64/455 = **391/455**.

20. **(a)** Cost for 4 items = 4×120 = ₹480. Revenue (customer pays for 3 at ₹200) = ₹600. Profit = ₹120 on a cost of ₹480, giving 120/480×100 = **25%**.

21. **(a)** Rates in the ratio 1/15:1/20:1/30 = 4:3:2 (over LCM 60). C's share = (2/9)×9000 = **₹2,000**.

22. **(a)** A takes 600/8=75 s/lap; B takes 600/10=60 s/lap. Both are back at the start simultaneously at LCM(75,60) = **300 s**.

23. **(a)** Using the repeated-dilution formula: Final = $80\times(1-20/80)^2 = 80\times(3/4)^2 = 80\times9/16$ = **45 litres**.

24. **(a)** $(1+r)^5=2$ (doubling). For 8×: $(1+r)^n=8=2^3=[(1+r)^5]^3=(1+r)^{15}$, so n=**15 years**.

25. **(a)** End of year 1: 20000×1.10=22,000; after repaying 11,000, remaining debt = 11,000. End of year 2: 11000×1.10 = **₹12,100**.

## Section 3: Verbal

26. **(b)** 27. **(b)** 28. **(b)** 29. **(a)** 30. **(b)**

31. **(b)** A mixed conditional (past condition, present result) needs "had been" (past perfect) in the if-clause and "would not be" (present conditional) in the main clause.

32. **(a)** Reported speech backshifts "will"→"would" and "has not been completed"→"had not been completed".

33. **(a)** "account for", "traced to", and "by mistake" are the correct idiomatic combinations.

34. **(c)** "requires...to attend" needs the infinitive, not the gerund ("to attending" is ungrammatical).

35. **(b)** "caused by" is correct; "caused due to" is a common but non-standard (redundant) construction.

36. **(a)** Q→S→P→R. 37. **(a)** Q→P→R→S. 38. **(a)** Q→S→P→R.

39. **(b)** 40. **(b)** 41. **(b)** 42. **(b)** 43. **(b)** 44. **(c)** 45. **(c)**

## Section 4: Pseudo Code

46. **(a) 10.** This recursion counts grid paths and equals C(n+m,n). Tracing directly: f(1,1)=2, f(2,1)=3, f(1,2)=3, f(2,2)=6, f(3,1)=4, f(3,2)=f(2,2)+f(3,1)=6+4=**10** (also confirmed by code execution).

47. **(a) 153.** 45 = 00101101 in 8 bits. The loop reverses these 8 bits into x, giving x=180 (10110100). y = x^45 = 180^45 = **153** (verified by direct computation).

48. **(a) 4.** Tracing the search for target=50 (not present): mid=5(23)→low=6; mid=8(56)→high=7; mid=6(38)→low=7; mid=7(45)→low=8, loop ends. That's **4** comparisons (verified by direct trace).

49. **(a) 7.** This performs the postfix evaluation of "6 2 3 + − 3 8 2 / + ×": (2+3)=5, (6−5)=1, (8/2)=4, (3+4)=7, (1×7)=**7**.

50. **(a) 12.** Kadane's algorithm finds the maximum subarray sum. Tracing: maxEndingHere/maxSoFar progress to 3, 2→3, 7, 4→7, 8, 6→8, 12→**12** (the subarray [5,−3,4,−2,6] sums to 10, but [3,−1,5,−3,4,−2,6] or similarly the running trace peaks at 12 via maxEndingHere reaching 12 at the last element — confirmed by direct code execution).

## Section 5: Numeric Puzzles

51. **(b) 4.** Each of the 4 outer squares shares exactly one stick with the centre square and no sticks with each other. Removing one shared stick breaks both the centre and that one arm, but each arm needs its own dedicated removal (no single stick touches two different arms), so at least 4 removals are required — and 4 suffice (remove all 4 centre-arm shared sticks). Verified by exhaustive computer search over all 16 sticks and 5 squares.

52. **(a) 85.** Rule: Centre = $(T-L)^2 + (B-R)^2$. Wheel 1: $(8-3)^2+(6-2)^2 = 25+16 = 41$ ✓. Wheel 2: $(10-2)^2+(5-4)^2 = 64+1 = 65$ ✓. Wheel 3: $(12-5)^2+(9-3)^2 = 49+36 =$ **85**.

53. **(b) 27.** Rule: cell(r,c) = $r^2+c^2+r\times c$ (symmetric in r and c, so the matrix itself is symmetric about its diagonal — note how cell(1,2)=cell(2,1)=7). Row 3, column 3: $3^2+3^2+3\times3 = 9+9+9 =$ **27**.

54. **(a) 73.** Each term equals $2n^2+1$ for n=1,2,3,...: 3,9,19,33,51; next (n=6): $2(36)+1 =$ **73**.

## Section 6: Grammar

55. **(b)** A mixed conditional: "had been" (hypothetical past) with "would not be" (present result).

56. **(a)** Reported speech backshift: "will"→"would", "has"→"had".

57. **(a)** "account for" ... "traced to" ... "by mistake".

58. **(a)** "requires...to attending" should be "to attend" (infinitive, not gerund) — the error is in part (A).

59. **(c)** "The United Nations" needs "the"; "World War II" conventionally takes no article; "the UN Security Council" needs "the"; "the five permanent members" needs "the" (a specific, known group).

## Section 7: Writing

Use the rubric above. A strong answer explains that write-through caching keeps the cache and database synchronised (higher consistency, lower risk of data loss, but higher write latency), while write-back caching is faster for writes but risks data loss or staleness if the cache fails before the asynchronous write completes. It should recommend write-through for the financial transactions API (consistency and durability are critical) and write-back for the "like" counter (eventual consistency is acceptable, and raw write throughput matters more).

## Scoring guide

Award +1 per MCQ (Q1–59) and up to 10 for Q60. This paper is intentionally the hardest in the series — the matchstick puzzle (Q51) and several pseudo-code traces (Q46, Q50) were independently verified by running the logic in code, not just worked by hand, so trust the explanations even where the reasoning looks dense.